Standard Operating Procedure (SOP) for Optical Transport Failure on IND-ACC-16 at Indiranagar Exchange using Arista CLI Dialect

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2937

Description

This SOP is designed to address optical transport failures on the Access Node (IND-ACC-16) at Indiranagar Exchange, which is a critical component of the telecom network. The failure domain in question pertains to the degradation of SFP/XFP modules, fiber cuts, and Rx/Tx power loss within the Arista network equipment. This scenario is a common occurrence due to the inherent nature of optical transport systems, where the signal degradation and attenuation over long-distance fiber optic cables can lead to issues such as signal quality degradation, packet loss, and ultimately, network downtime. The Access Node, being a critical point of entry for multiple services, is particularly susceptible to these types of failures. The Arista CLI dialect is utilized throughout this SOP to ensure compatibility with the vendor-specific equipment. It is essential to note that the resolution of this failure domain requires a thorough understanding of Arista's EOS (Extensible Operating System) and its underlying protocols, including but not limited to, OSPF, BGP, and STP.

Pre-requisite Checks

- Check: Verify the site location and node ID (IND-ACC-16) to ensure accuracy.

- Check: Confirm the presence of redundant links and verify their status to ensure continuous network operation.

- Check: Dispatch remote hands to the Indiranagar Exchange to facilitate physical access to the node in case of a hardware-related failure.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss due to Optical Attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: [list of interfaces, including any errors or warnings], Description: Run this command to verify the status of all interfaces, including any errors or warnings that may indicate an optical transport failure.
```

```
Command: show optics, Expected Output: Optics Details: [list of optic modules, including temperature, voltage, and current readings], Description: Execute this command to retrieve detailed information about the optic modules, including temperature, voltage, and current readings, which can help identify potential issues.

Command: show lldp neighbors, Expected Output: LLDP Neighbors: [list of neighboring devices, including their MAC addresses and port numbers], Description: Run this command to verify the presence and status of neighboring devices using LLDP (Link Layer Discovery Protocol).
```

Resolution Steps

Step 1: Step: Verify SFP/XFP Module Status, Command: show optics, Description: Execute the show optics command to verify the status of the SFP/XFP modules and identify any issues with temperature, voltage, or current readings. If issues are detected, proceed to the next step.

Step 2: Step: Replace SFP/XFP Module (if necessary), Command: replace optics, Description: If the SFP/XFP module is faulty, execute the replace optics command to remove and replace the defective module with a new one. Ensure that the new module is properly seated and secured before proceeding.

Step 3: Step: Verify Fiber Connection, Command: show interface status, Description: Run the show interface status command to verify that the fiber connection is secure and functioning properly. If issues are detected, proceed to the next step.

Step 4: Step: Verify Rx/Tx Power Levels, Command: show optics, Description: Execute the show optics command to verify that the Rx/Tx power levels are within the acceptable range. If issues are detected, proceed to the next step.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status: [list of interfaces, with no errors or warnings], Description: Verify that the interface status is operating within normal parameters, with no errors or warnings.

Command: show optics, Expected Output: Optics Details: [list of optic modules, with normal temperature, voltage, and current readings], Description: Verify that the optic modules are functioning within normal parameters, with no issues detected.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Module, Command: show optics, Description: If the replacement module is not functioning properly, restore the original SFP/XFP module and verify that the issue is resolved.

- Step: Re-establish Fiber Connection, Command: show interface status, Description: If the fiber connection is not secure, re-establish the connection and verify that the issue is resolved.